School of Computing and Augmented Intelligence
IRA A. FULTON SCHOOLS OF engineering

CSE 446/598

Software Integration and Engineering

Unit 1

Introduction and Advanced Service Development

Lecture 1-2
Raw Services and Self-Hosting

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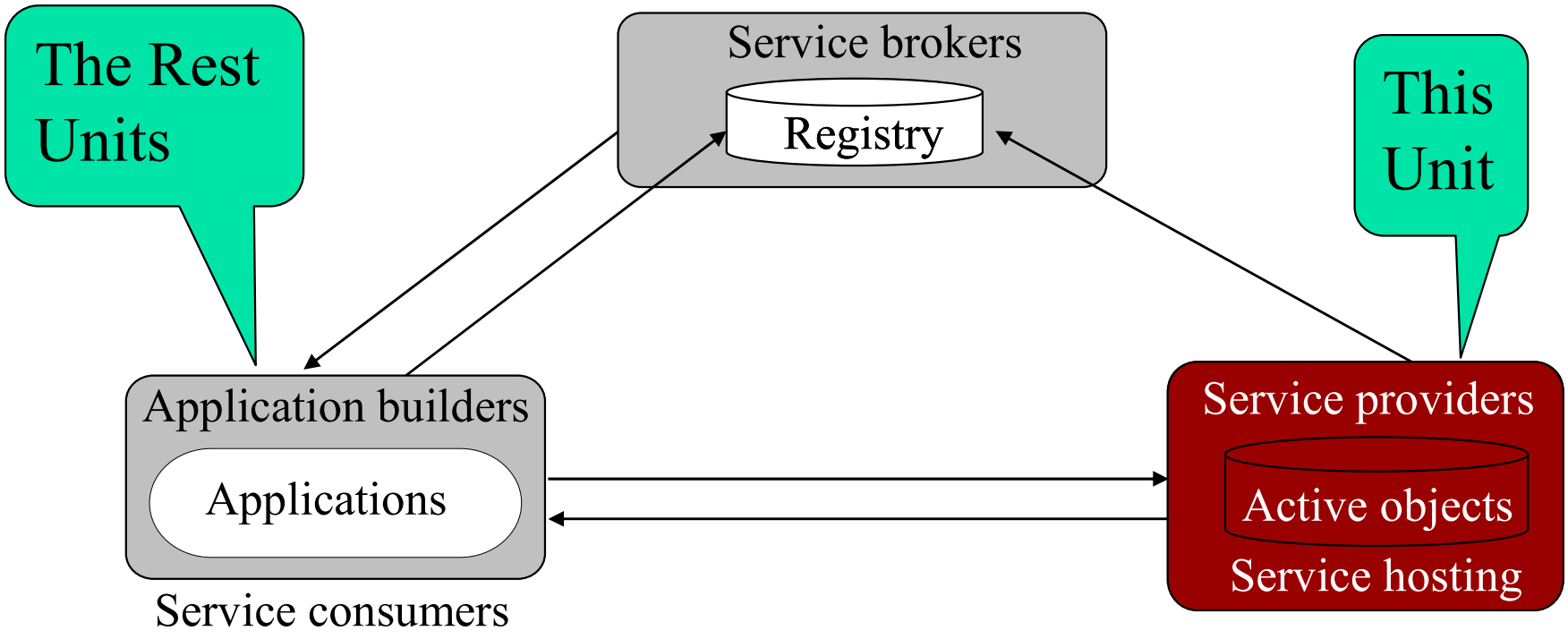
https://myasucourses.asu.edu/

Lecture Overview

- WCF Basic Service Development and Consuming (Review)
- Service is an active API with language dependent or independent interface;
- API can be active (hosted) or passive, and it can have language dependent or independent interface.
- API example: Raw Service Development and Self-Hosting
  - Raw Service development: Develop the interface without using the service development tools
  - Host (worker process) development: Make it active
  - Endpoint creation
  - Proxy creation
- Different Clients Consuming Self-Hosting Services

Model of Service-Oriented Software Development

Three-Party Development Model



Service/Application Development Environments

- **ASP.NET and Windows Communication Foundation (WCF)** (Text Chapter 3)
- Service development in Python/Flask
- Workflow Foundation (Text Chapter 7) and Power Platform (New)
- Oracle SOA Suite/ JDeveloper (Text Chapter 8)
- AI-assisted IDEs: Power Platform and Antigravity
- Java-based SOA Development: Netbeans, Eclipse/Axis/Tomcat
- JAX-RPC 1.1 API
- J2EE
- BEA WebLogic
- IBM WebSphere
- SAP Enterprise Service Architecture (ESA)

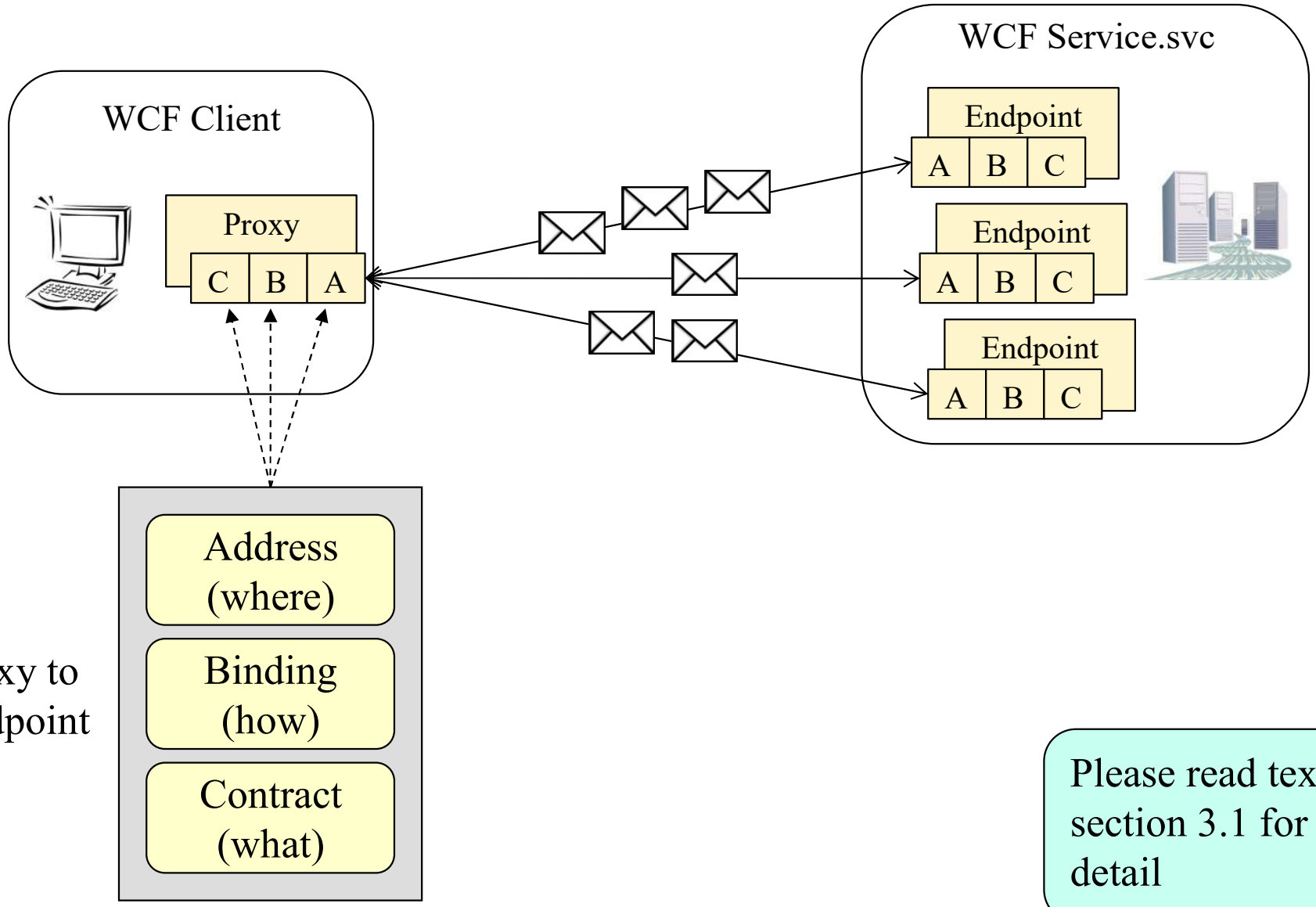
WCF Service Development and Consuming

- ASP .Net framework is a component in Visual Studio for service and application development.
  - It can be used for service development (.asmx service)
  - It is mainly used for application development
- WCF is a component in Visual Studio
  - WCF is used for service development only (.svc service)
  - It is used for developing SOAP as well as RESTful services
  - WCF service hosting: with or without IIS
- Service Interfaces: Both .asmx and .svc services use WSDL
- WCF service consuming
  - Console application
  - ASP .Net Windows forms application (Desktop application)
  - ASP .Net Website (Web application) / HTML5 Website
  - BPEL, and many others

Different at language interface level

Same at service interface level

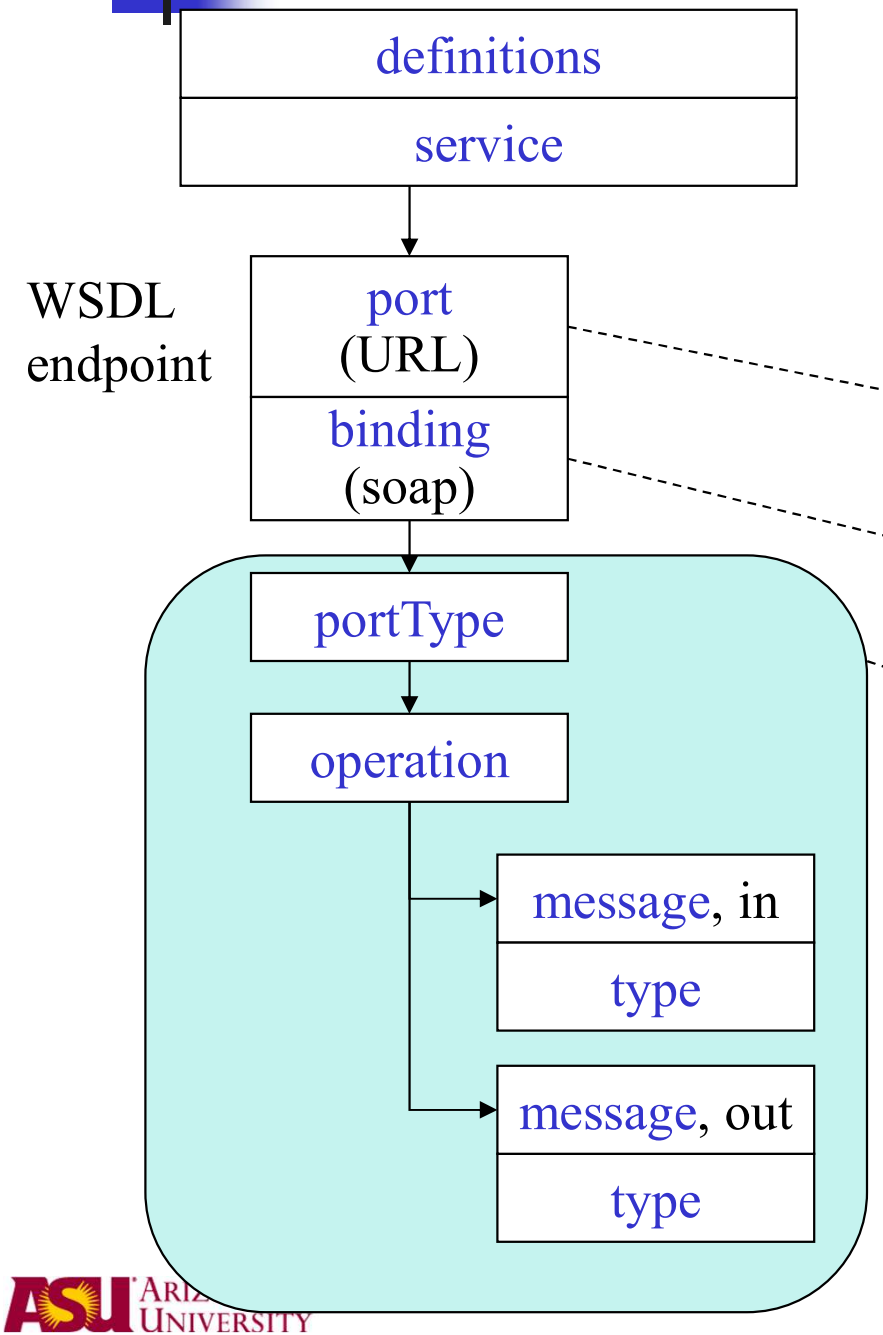
WCF Interface between Client and Service



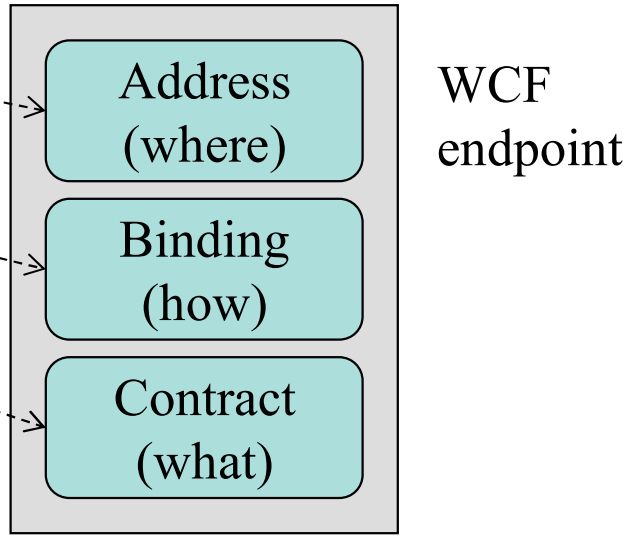
Proxy to Endpoint

Please read text section 3.1 for more detail

Endpoint Coded in WSDL



- An ASP .Net endpoint is the port to the method exposed through a markup of contract in program
- WCF endpoint contains ABC:

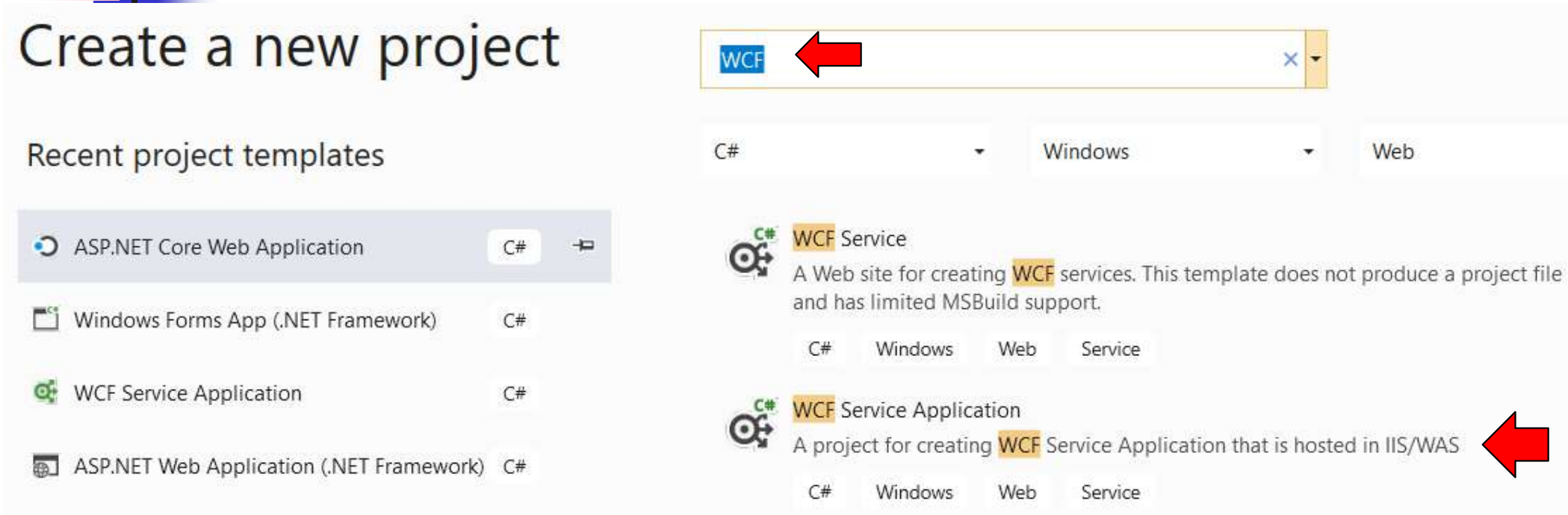


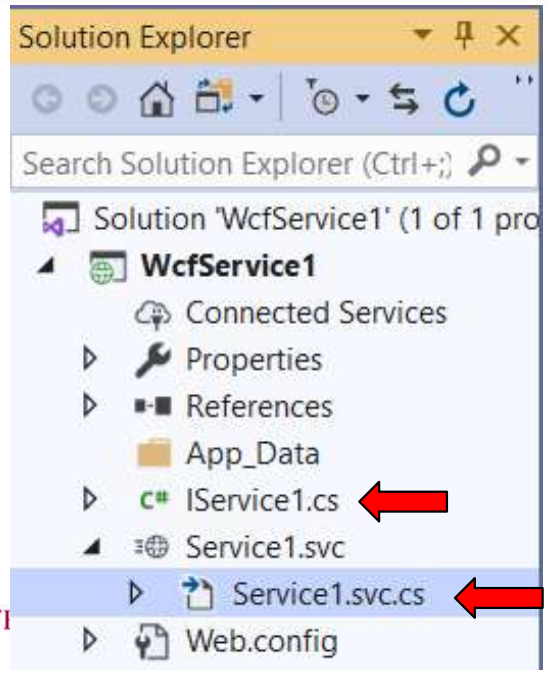
- Different development tools (.asmx, .svc, and others) have different structures, but contain the same basic information
  - All are coded in WSDL

Developing WCF Basic Services & Clients

- Developing a Service using WCF tools (Review)
  - Developing a WCF service using template
  - Hosting the service using IIS Express Server
  - Developing a console application to consume the WCF service
- Developing WCF Service with Self-Hosting

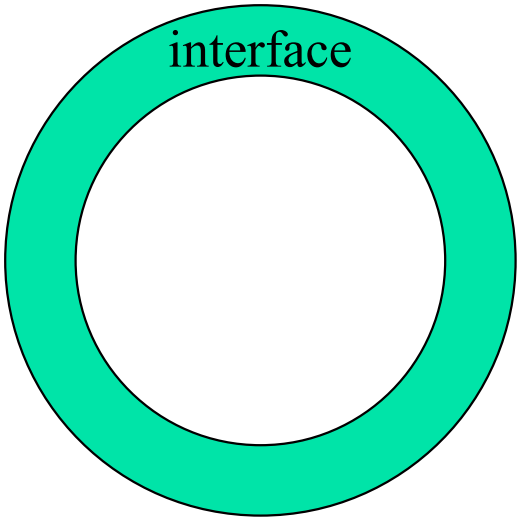
Develop Web Services Using WCF





Text Chapter 3, section 3.2.1
We did this exercise in CSE445 Assignment 1

Step 2: Define Service Contract / Interface




```
// IService.cs file
using System.ServiceModel;

[ServiceContract]
public interface IService {
    [OperationContract]
    string GetData(int value);    // generated code, rename it
    [OperationContract]
    double PiValue();             // my operation 1
    [OperationContract]
    int absValue(int intVal);     // my operation 2
}
```

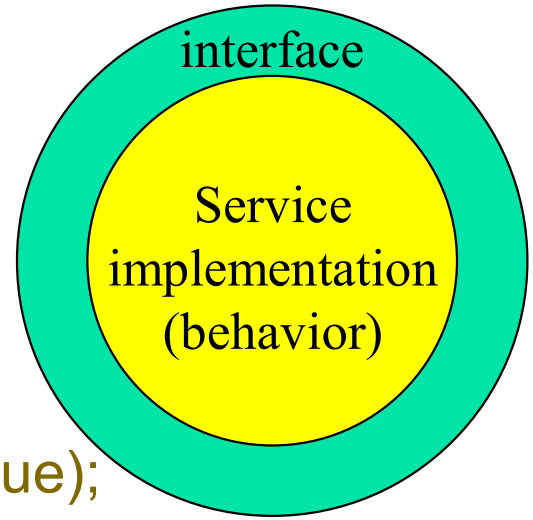
1

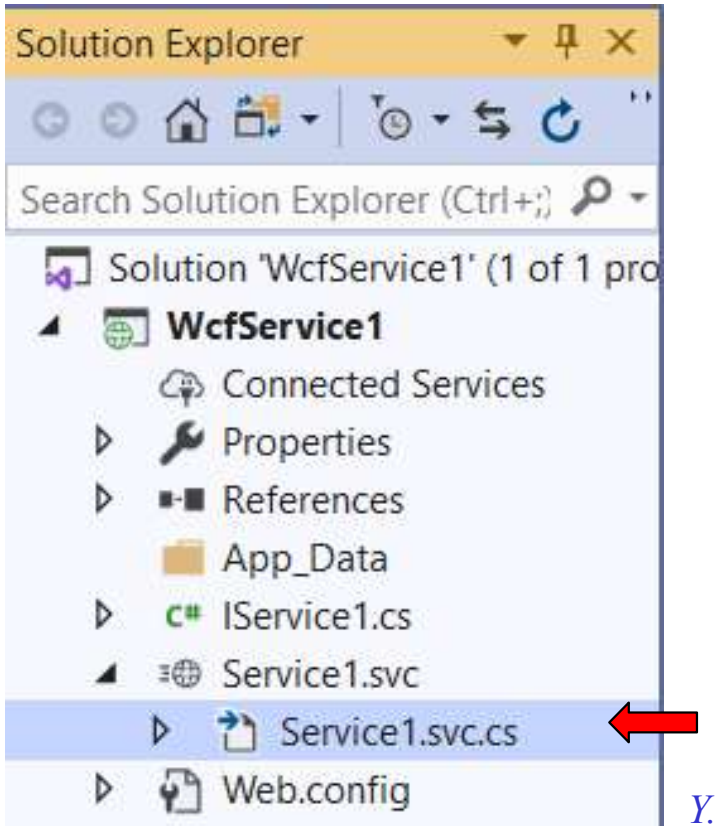
Step 3: Implement Service Contract (behavior)

2

```
// Service.svc.cs file
using System.ServiceModel;

public class Service : IService {
    public string GetData(int value) {
        return string.Format("You entered: {0}", value);
    }
    public double PiValue() {
        double pi = System.Math.PI;
            return (pi);
    }
    public int absValue(int x) {
        if (x >= 0) return (x);
            else return (-x);
    }
}
```





Step 4: Build and View in Browser

Right-click file Service.svc and choose "View in Browser"

Blackboard Learn | My ASU - Roster | Service Service

localhost:57956/BasicThreeSvc/Service.svc

Yinong Chen | Bookmark Manager | Other bookmarks

Service Service

The service is hosted on localhost / IIS Express

You have created a service.

To test this service, you will need to create a client and use it to call the service. You can do this using the svcutil.exe tool from the command line with the following syntax:

```
svcutil.exe http://localhost:57956/BasicThreeSvc/Service.svc?wsdl
```

This will generate a configuration file and a code file that contains the client class. Add the two files your client application and use the generated client class to call the Service. For example:

C#

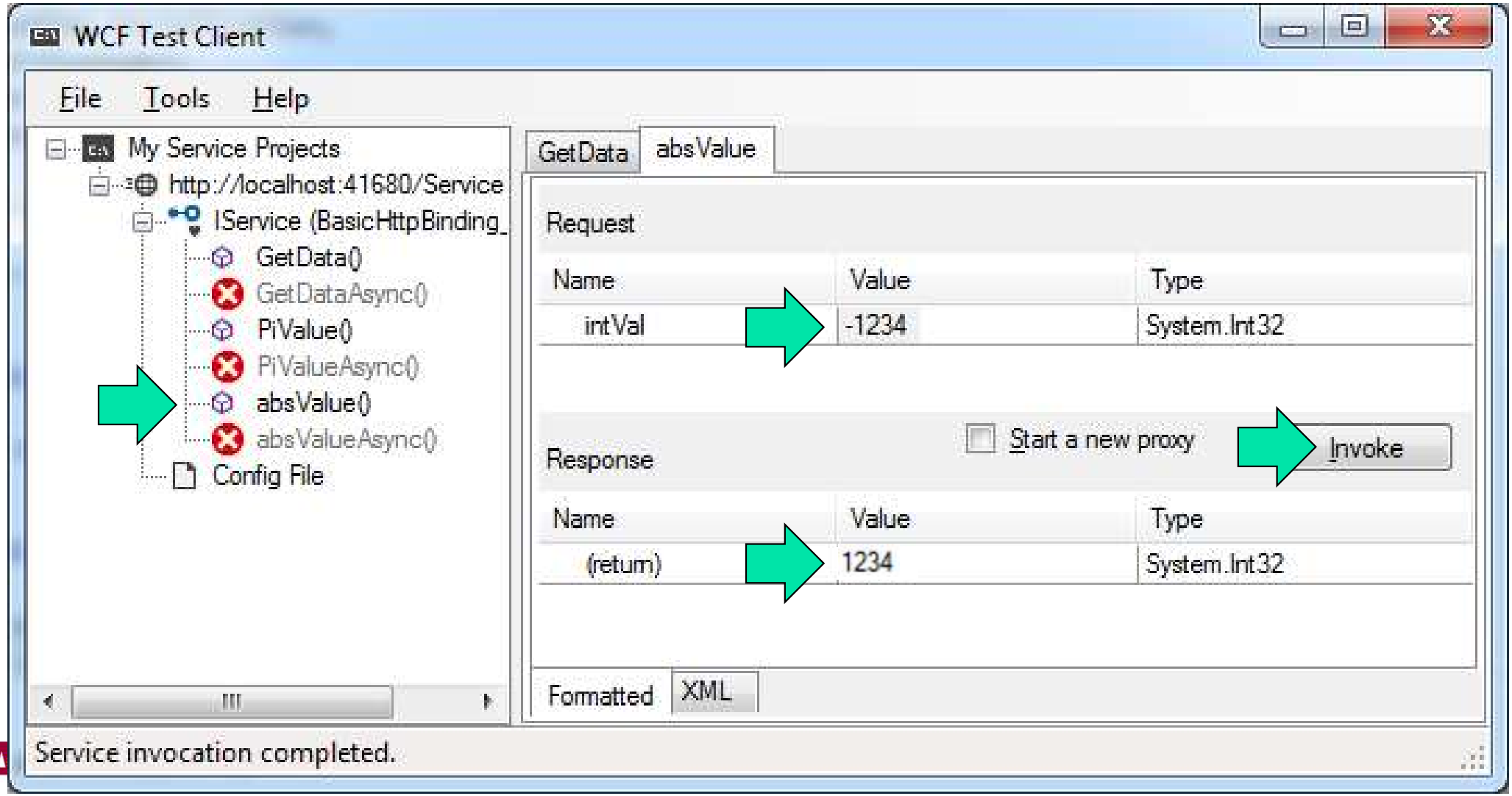
```
class Test
{
    static void Main()
```

Click on the link:
http://localhost:57956/BasicThreeSvc/Service.svc?wsdl
To view the WSDL file

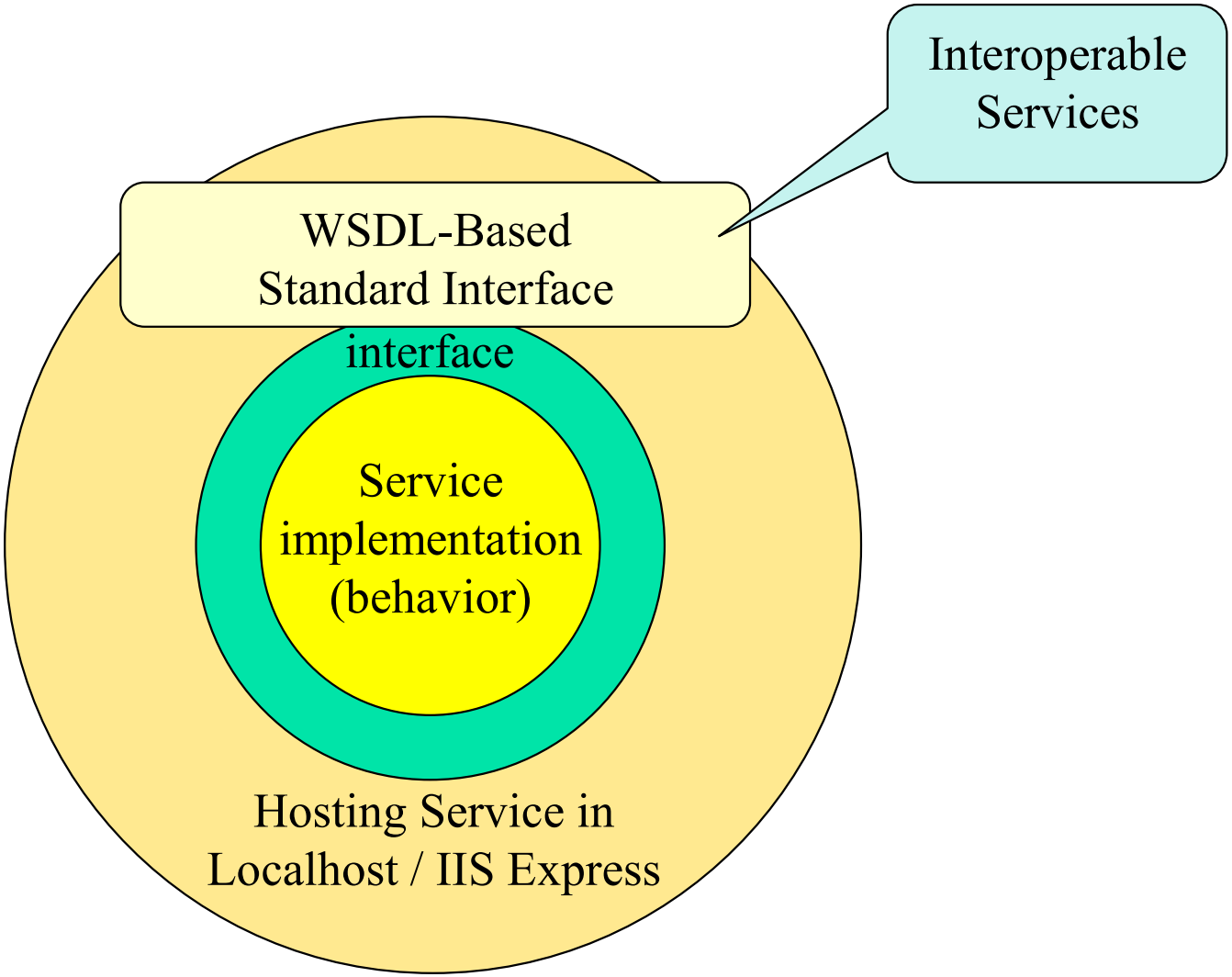
Built-in Test Client

Select the service
Menu: DEBUG → Start Without Debugging
The Test Client Window will be open:

Read text for other ways of starting the Test Client: From CMD

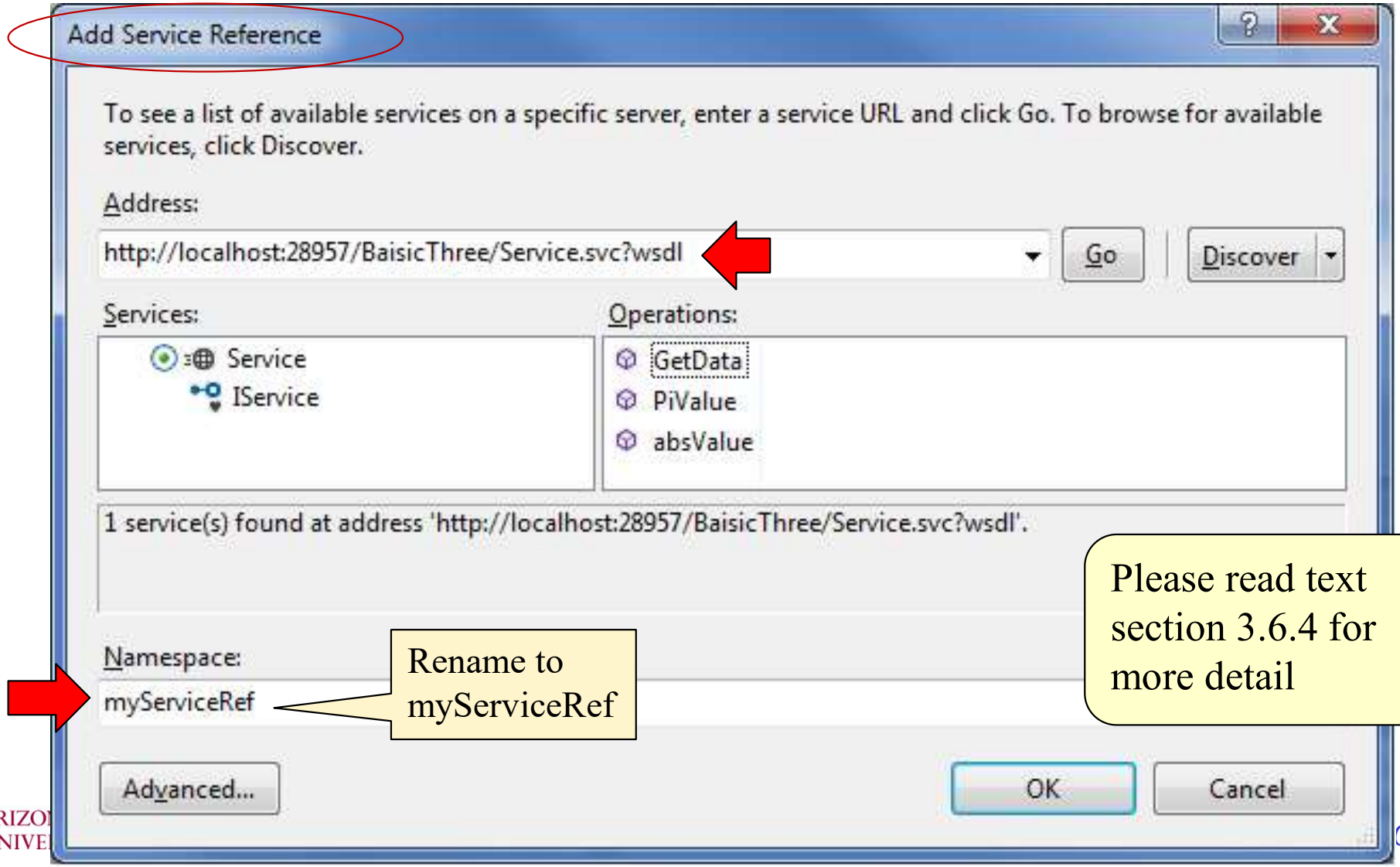


Hosting Using Localhost / IIS Express



Step 5: Create a Client to Test the Service

- Add a Console Application into the same solution
- Add Service Reference: `http://localhost:28957/BaisicThree/Service.svc?wsdl`



Rename to myServiceRef

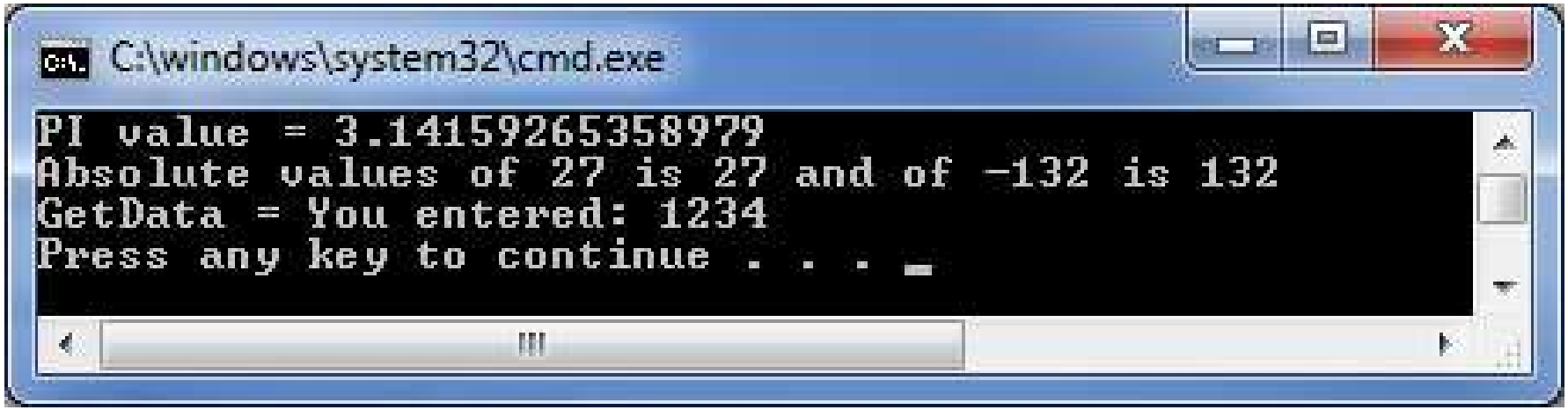
Please read text section 3.6.4 for more detail

Step 6: Write Client Code to Invoke Service

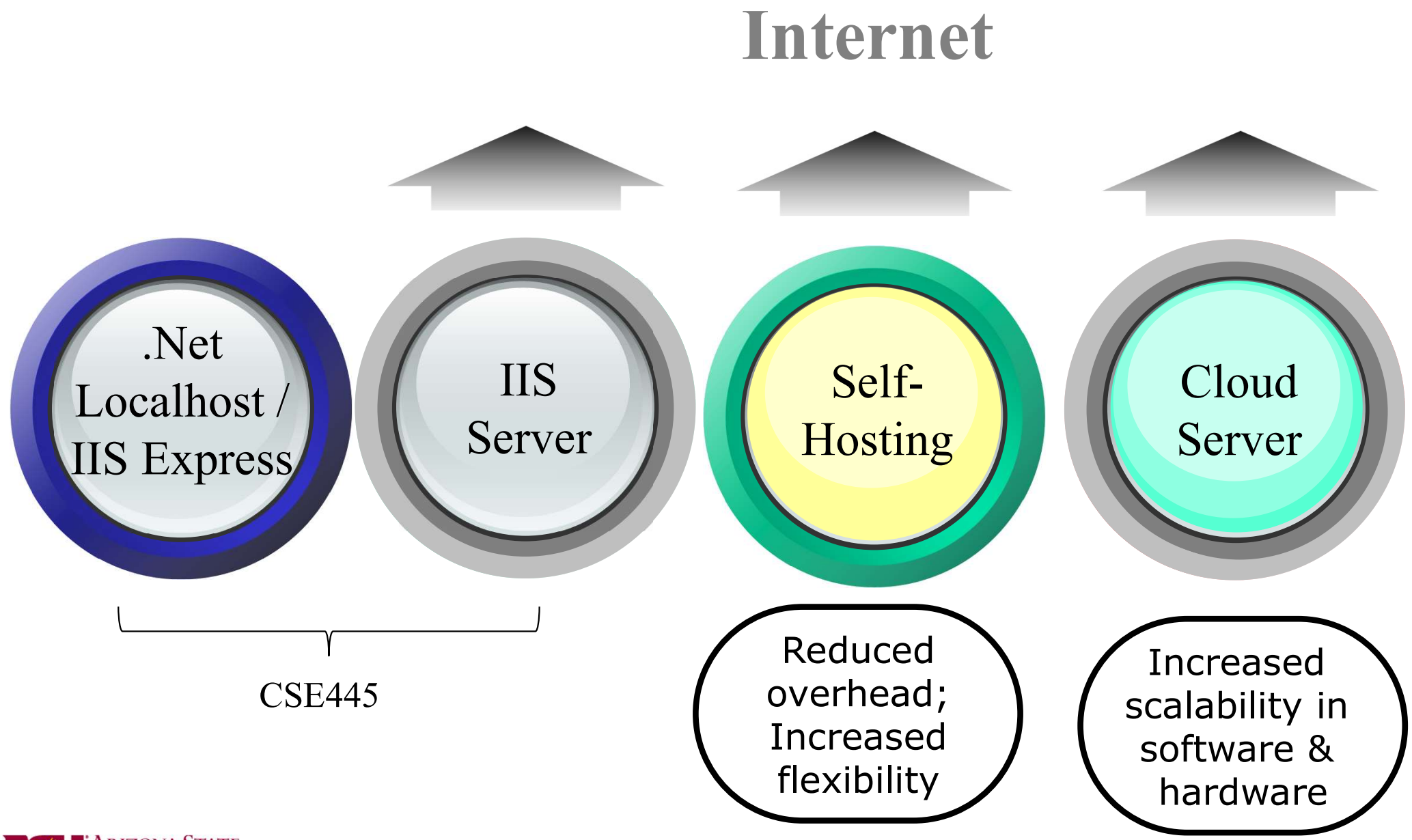
```
static void Main(string[] args) {   // console application
    myServiceRef.ServiceClient myProxy = new myServiceRef.ServiceClient();
    double pi = myProxy.PiValue();  // call PiValue operation
    Int32 test1 = 27;   // Create test input 1
    Int32 test2 = -132; // Create test input 2
    Int32 result1 = myProxy.absValue(test1); // call operation
    Int32 result2 = myProxy.absValue(test2); // call operation
    Console.WriteLine("PI value = {0}", pi);
    Console.WriteLine("Absolute values of {0} is {1} and of {2} is {3}", test1,
result1, test2, result2);
    Console.WriteLine("GetData = {0}", myProxy.GetData(1234));
    // Close the proxy, the channel to the service
    myProxy.Close();
}
```

Step 7: Test the WCF Client and Service

- First, start the service by right-click the **Service.svc** and choose "View in Browser" → Keep service running
- Right-click the **client** project and choose: Set as StartUp Project
- Start without Debugging



Different Ways of Hosting WCF Services



Developing Self-Hosting API and Clients

- WCF Basic Service Development (Review)
  - The process is largely automated
- Developing an API with Self-Hosting step by step
  - Developing API without using service template tools;
  - Developing the hosting service;
  - Creating endpoint and proxy
  - Consuming the API in a command line client;
  - Consuming the API in ASP .Net website

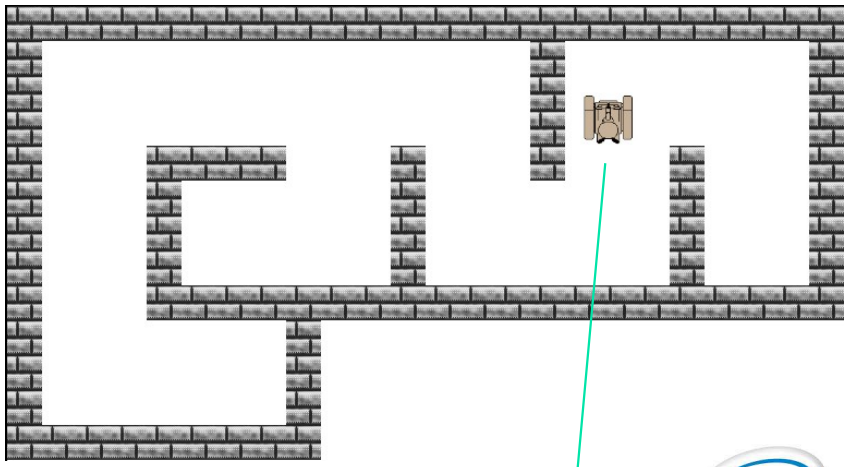
Reasons of Using Self-Hosting

- Reduced overhead for efficiency
  - IIS (Internet Information Services) and WAS (Windows Process Activation Services) are designed to deal with all possible scenarios of service hosting (https://docs.microsoft.com/en-us/dotnet/framework/wcf/feature-details/hosting-in-windows-process-activation-service)
  - Service interface and service protocol can increase overhead
  - Self-hosting can avoid many overheads
- Increased flexibility/controllability
  - IIS and WAS decide when a service starts and terminates.
  - With self-hosting, you can decide when to start and when to close a service.

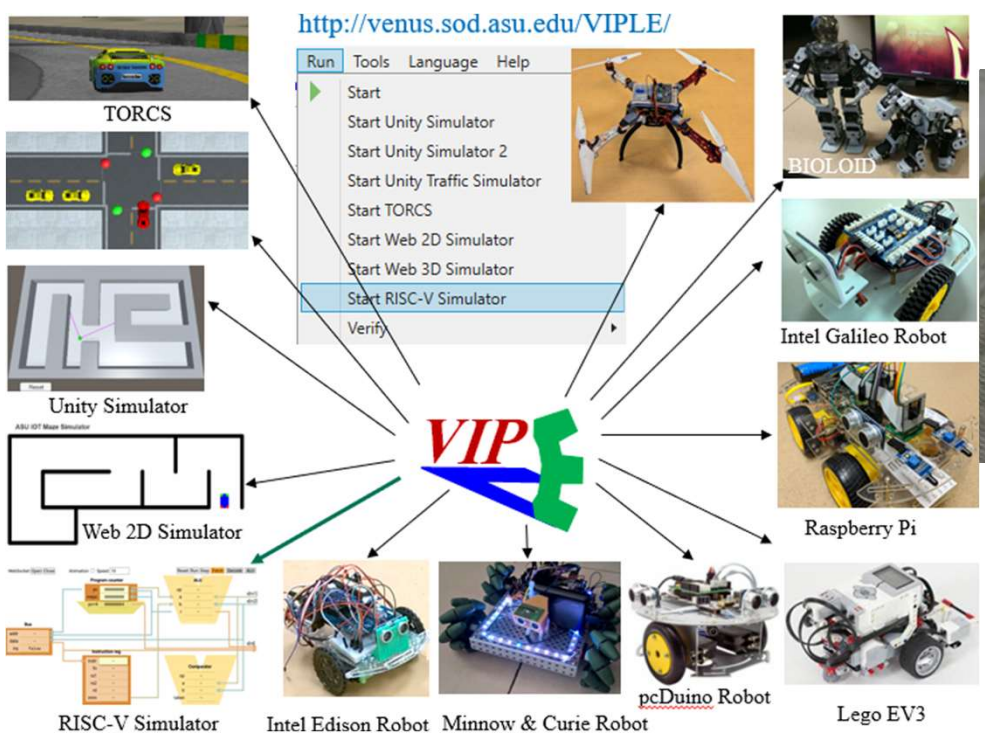
Case Study of Using Self-Hosting Service for IoT and Robotics Apps

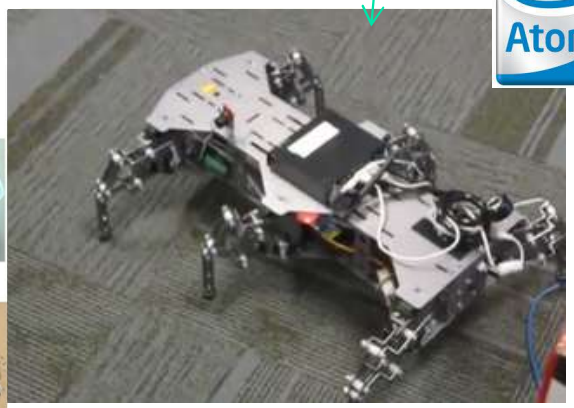
A Motivation for Self-Hosting

Web Simulation Environment



Visual **I**oT/Robotics
Programming **L**anguage
Environment





Robot as a Service at ASU, with self-hosting service

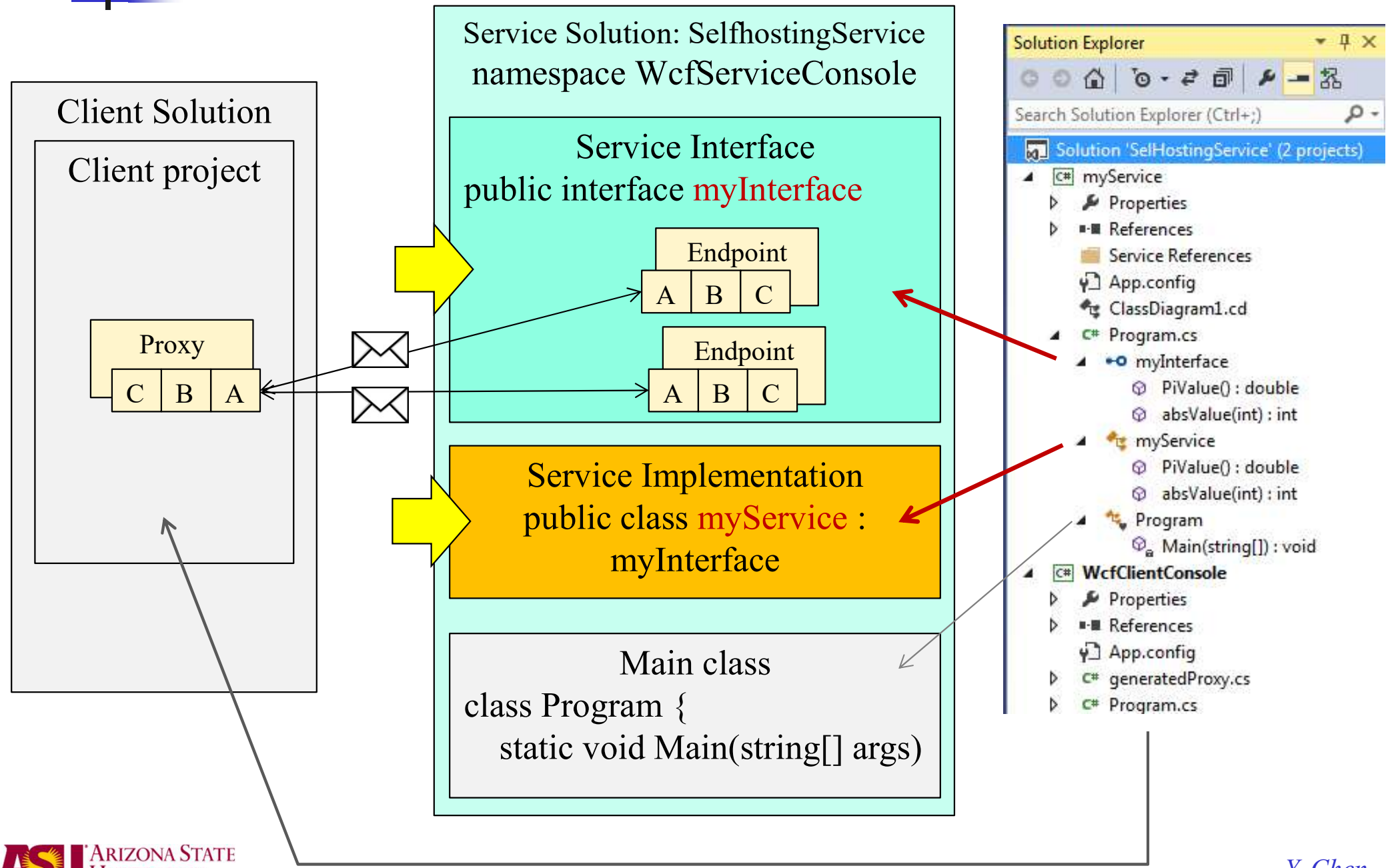
ASU Arizona State University

Y. Chen

Self-Hosting Using Multithreading

- Write your own Worker process, which can run in an infinite loop, or until a condition is met;
- Write your service methods
- Worker process (Main) activates each service method as a thread – active object (design pattern)
- Without implementing service management, one client can call the service at a time;
- To implement multiple accesses at the same time, implement service queue, as discussed in Text Chapter 2.

Self-Hosting Project Overview



Developing Services with Self-Hosting

Reading: Text Section 7.1

1. Start Visual Studio as an administrator: right-clicking the program in the Start menu and selecting "Run as administrator"; It is necessary as the service needs to register an HTTP channel for client to access;
2. Create a new Solution, with a new Project of Console Application type and name the service "myService";
3. Create Contract Through Interface;
4. Implement the service behavior in C#
5. **Add Hosting Service and Endpoint**

Service Implementation
public class myService : myInterface

Starting a New Solution of Console Application Type

We use console application to create a service

Create a new project

console

Recent project templates

C#

Windows

Windows Forms App (.NET Framework) C#

Console App (.NET Framework)

A project for creating a command-line application

C# Windows Console

Configure your new project

Console Application (.NET Framework) F# Windows Console

Project name

ServiceConsole

Location

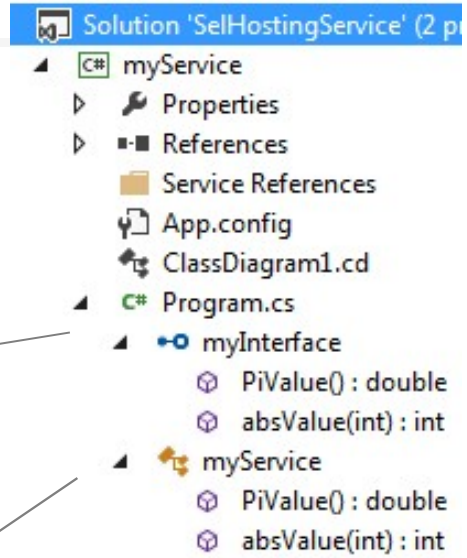
C:\Users\ychen\Dropbox (ASU)\Current in Dropbox\Teach All\CSE445-446 Programs

Console application is a basic template without advanced tools

Service Code in the Console Application: Interface and Implementation

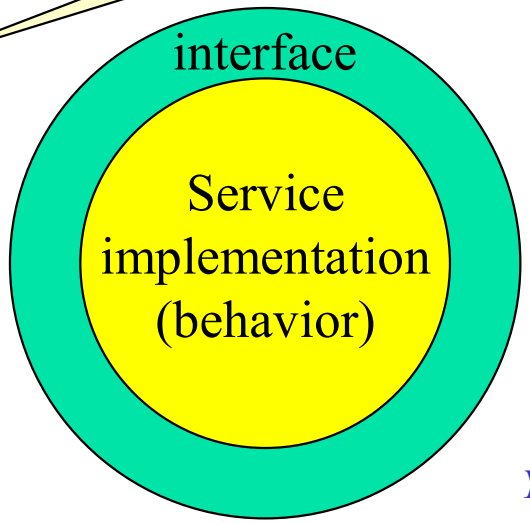
```
using System;
using System.ServiceModel;
using System.ServiceModel.Description;
namespace ServiceConsole {
    [ServiceContract]
    public interface myInterface { // Use interface to define contracts
        [OperationContract] double PiValue();
        [OperationContract] int absValue(int intVal);
    }
    public class myService : myInterface  // implementation of interface
    {     public double PiValue() {
                double pi = System.Math.PI;  return (pi);
          }
          public int absValue(int x) {
                if (x >= 0) return (x); else return (-x);
          }
    } // not yet complete, to add hosting code next
```

Service Contract is based on interface definition

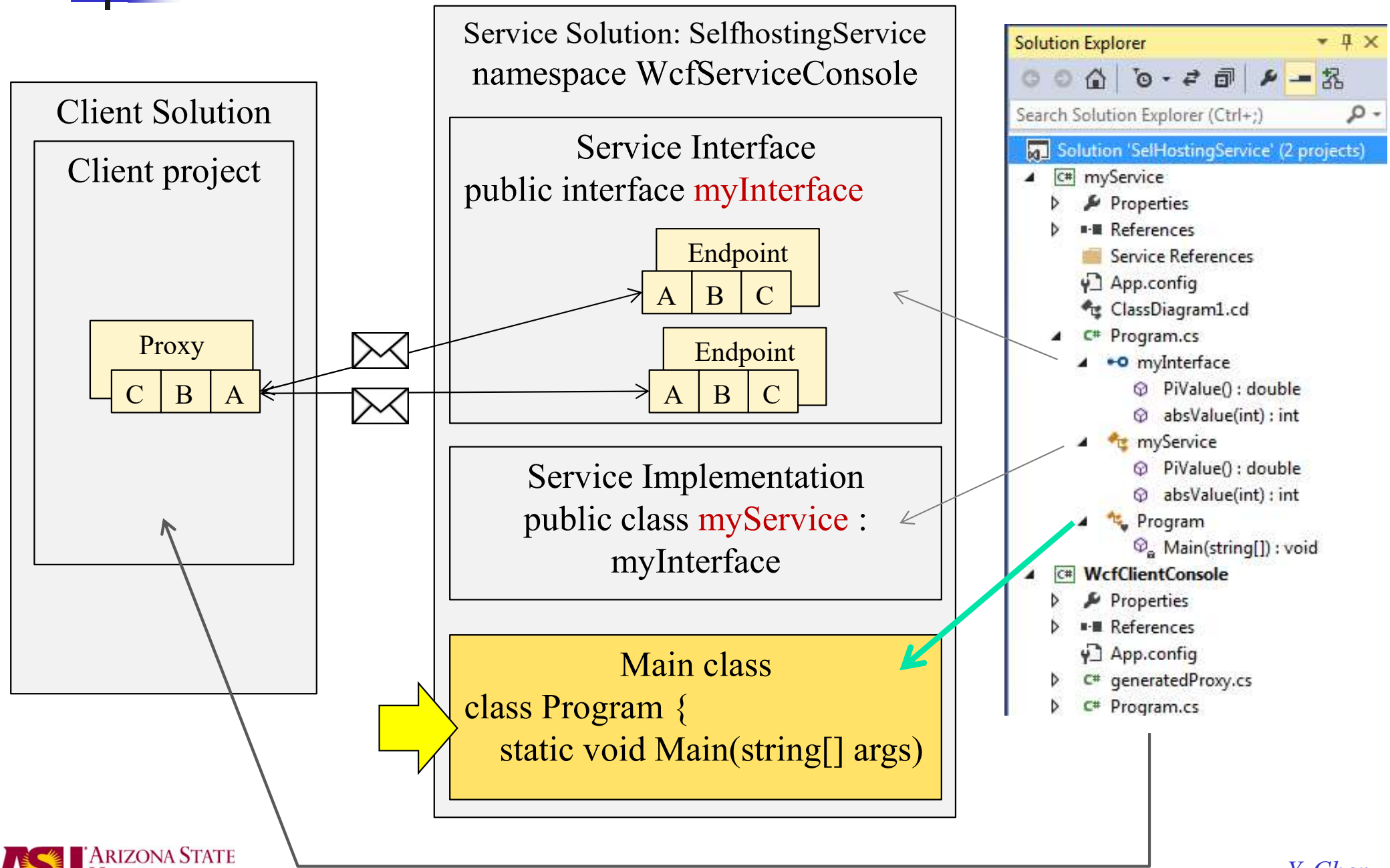


Operation Contract is based on method

Implementation



Project Overview



Create Hosting Service and Endpoint

In the same program, add a class with the Main method;

1. Using Uri library class to create a URI instance to myService as the base address;

```
Uri baseAddress = new
    Uri("http://localhost:8000/Service");
```

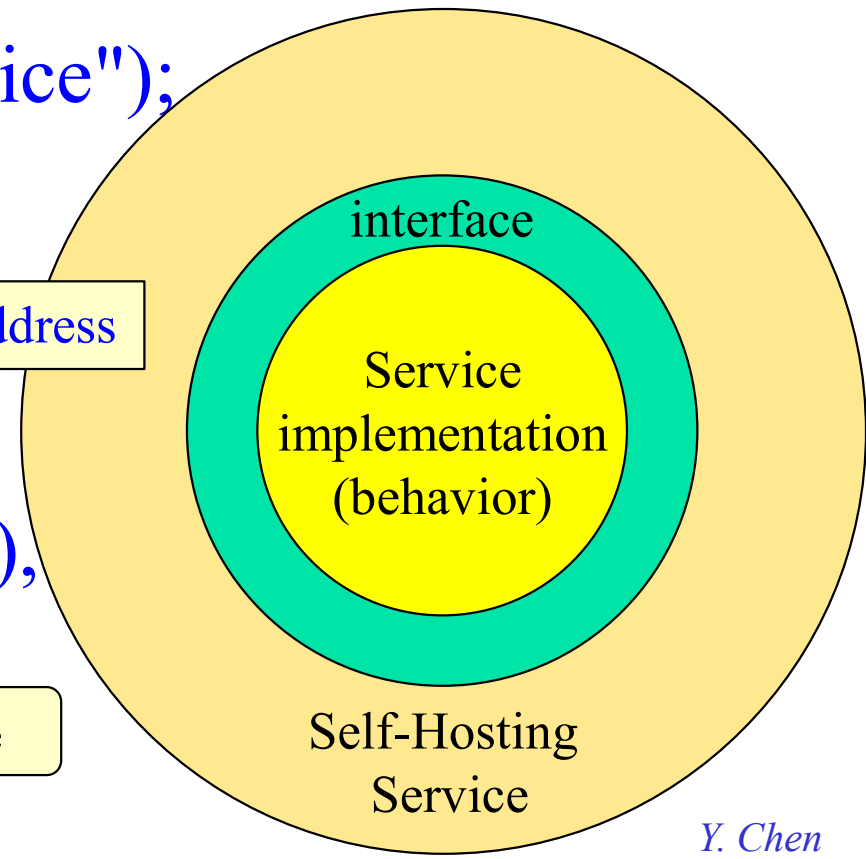
2. Create a new ServiceHost instance to host the service

```
ServiceHost selfHost = new
    ServiceHost(typeof(myService),
    baseAddress);
```

Address

Service name

Base address included



Create Hosting Service and Endpoint

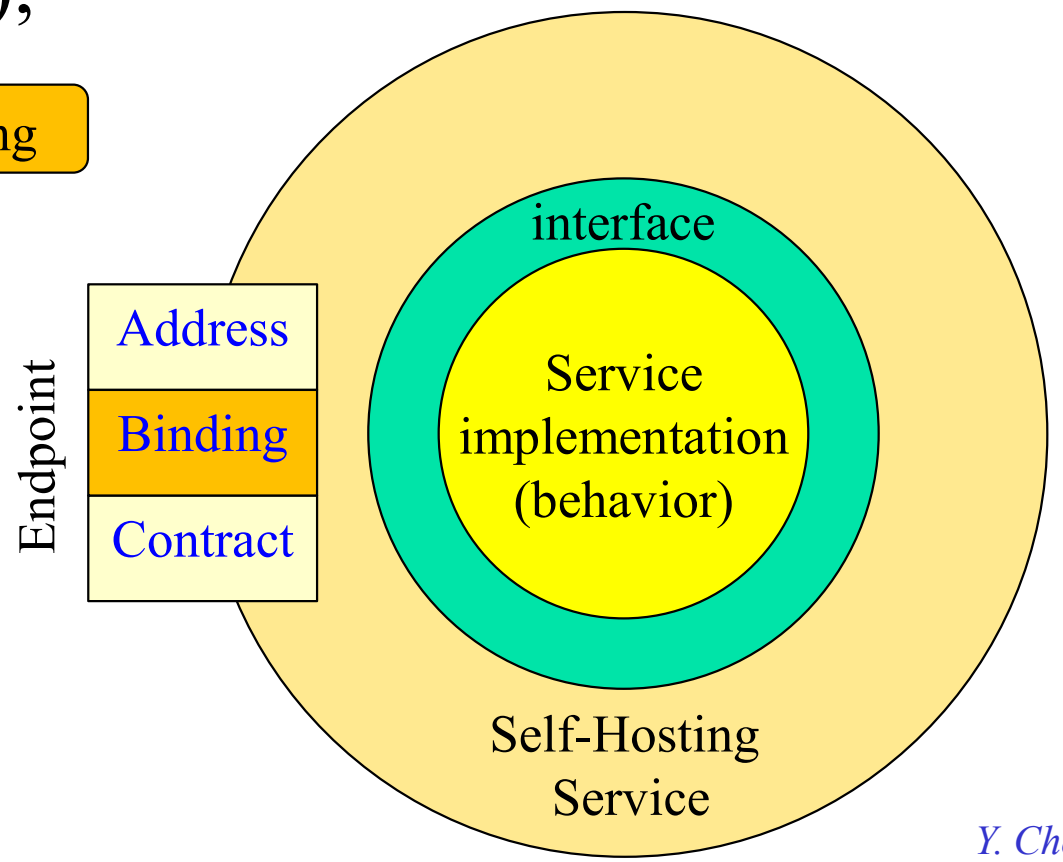
3. Add contract (myInterface) and Binding (WsHttpBinding)

```
selfHost.AddServiceEndpoint(typeof(myInterface),
    new WsHttpBinding(),
    "myService");
```

Contract

binding

behavior



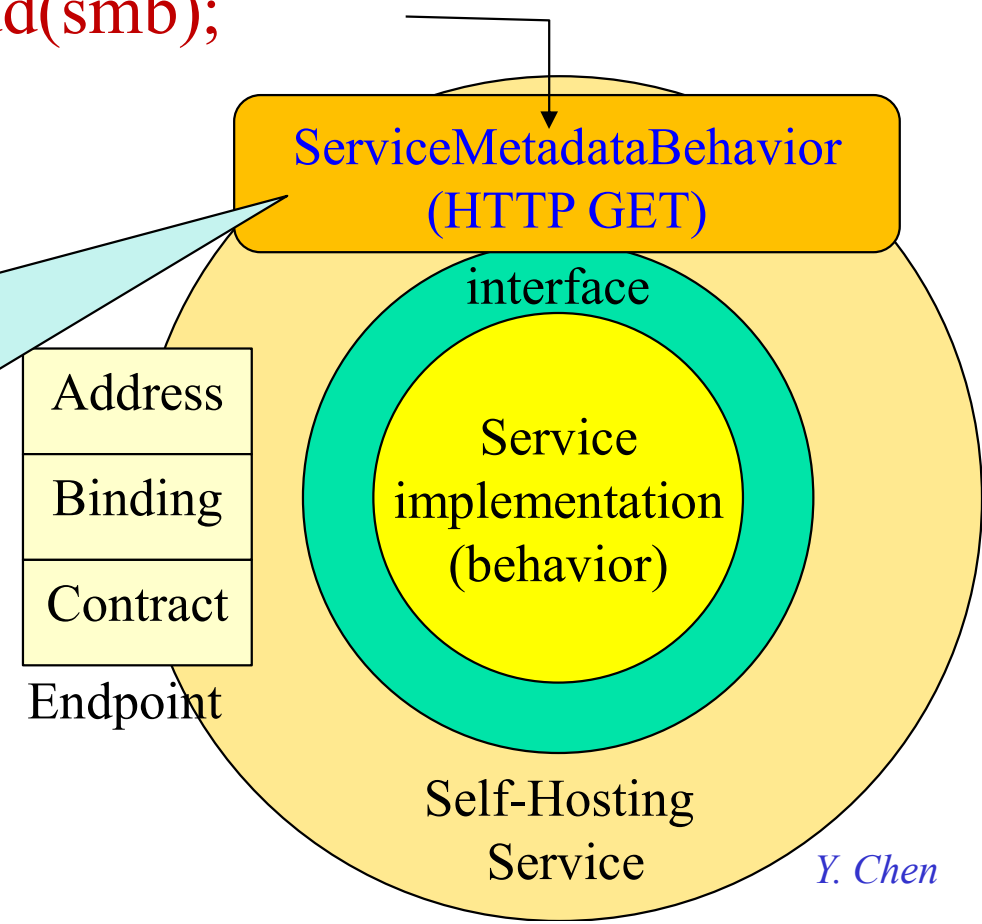
Adding HTTP GET for Public Access

4. Add metadata for public access

System.ServiceModel.Description.ServiceMetadataBehavior

```
smb = new System.ServiceModel.Description.ServiceMetadataBehavior( );
smb.HttpGetEnabled = true;
selfHost.Description.Behaviors.Add(smb);
```

A different class in the namespace System.ServiceModel.Description will result in a different interface for a different kind of clients to access. As both SOAP-WSDL services and RESTful services use HTTP GET, the interface can potentially support both types of services.



Public Properties of ServiceMetadataBehavior Class

Property Name	Description
ExternalMetadataLocation	Gets or sets a value that is the location of service metadata
HttpGetEnabled	Gets or sets a value that indicates whether to publish service metadata for retrieval using an HTTP/GET request.
HttpGetUrl	Gets or sets the location of metadata publication for HTTP/GET requests.
HttpsGetEnabled	Gets or sets a value that indicates whether to publish service metadata for retrieval using an HTTPS/GET request.
HttpsGetUrl	Gets or sets the location of metadata publication for HTTPS/GET requests.
MetadataExporter	Gets or sets the internal MetadataExporter object used to publish the service metadata.

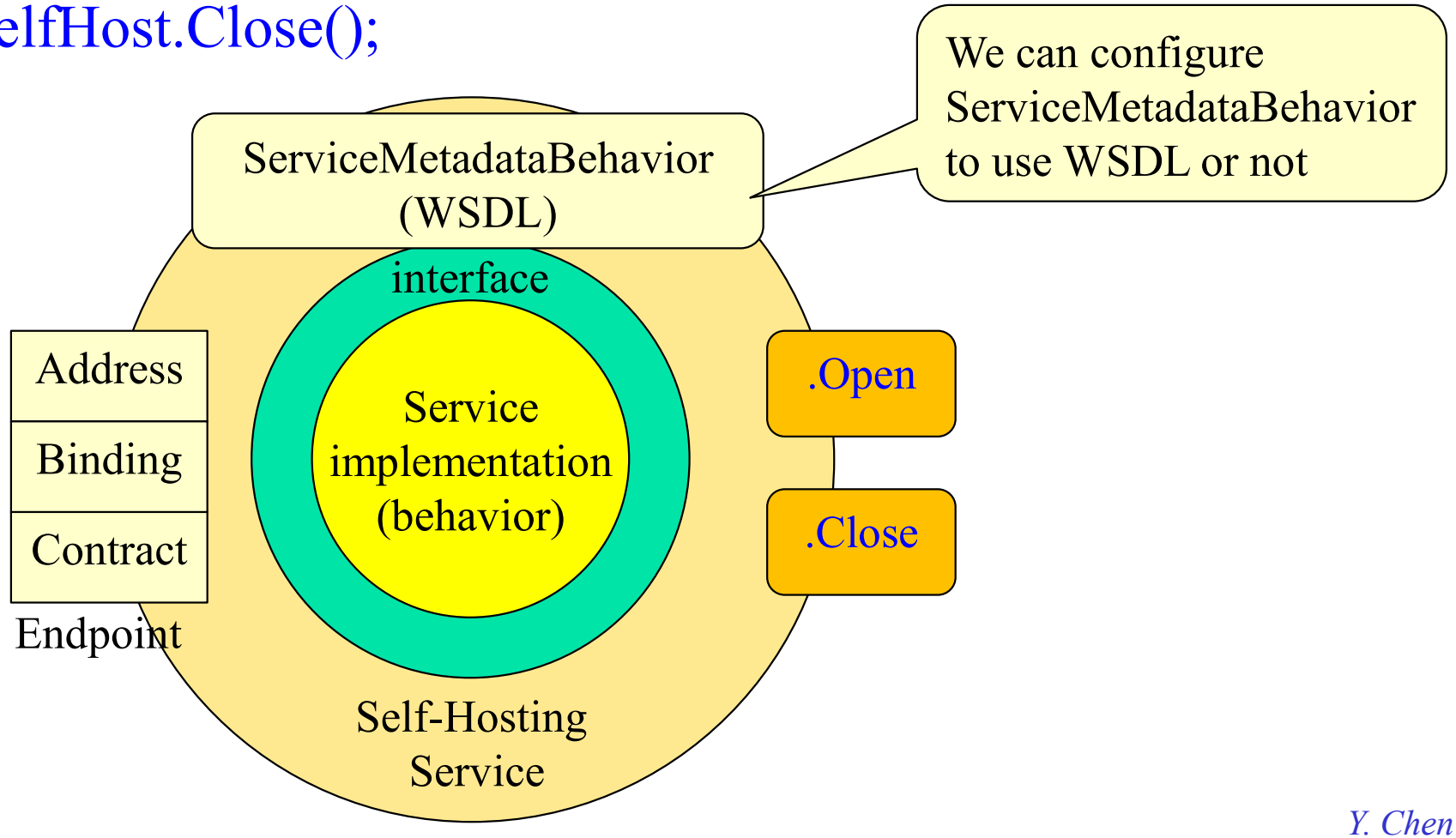
Starting and Terminating the Service

5. Start service and waiting for request.

```
selfHost.Open();
```

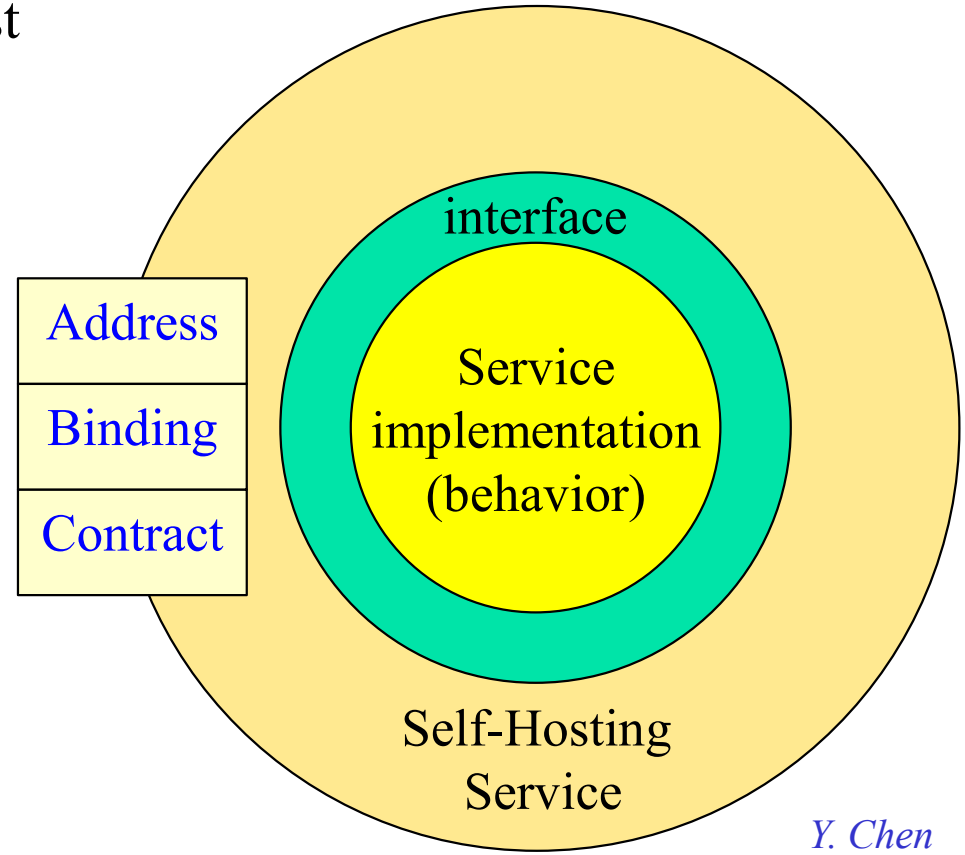
6. Close the ServiceHostBase to shutdown the service.

```
selfHost.Close();
```



Code Together: Hosting Part

```
class Program { // more detail see text section 7.1
    static void Main(string[] args) {
        // (1) Create a URI instance to myService as the base address.
        Uri baseAddress = new Uri("http://localhost:8000/Service");
        // (2) Create a new ServiceHost instance to host the service
        ServiceHost selfHost = new ServiceHost
                (typeof(myService),
                baseAddress);
        try {
        // (3) Add contract
            selfHost.AddServiceEndpoint(
                typeof(myInterface),
                new WSHttpBinding(),
                "myService");
```



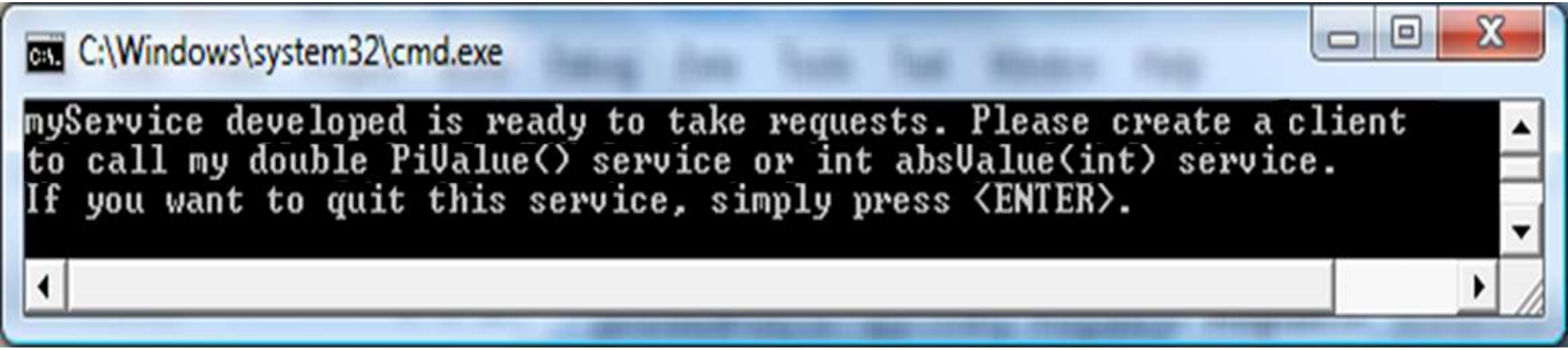
Service Code: Hosting Part

```
// (4) Add metadata for public access.
System.ServiceModel.Description.ServiceMetadataBehavior smb =
      new  System.ServiceModel.Description.ServiceMetadataBehavior( );
       smb.HttpGetEnabled = true;
       selfHost.Description.Behaviors.Add(smb);
// (5) Start the service and waiting for request.
selfHost.Open();
Console.WriteLine("myService developed is ready to take requests. Please
create a client to call my double PiValue() service or int absValue(int)
service.");
Console.WriteLine("If you want to quit this service, simply press
<ENTER>.\n");
Console.ReadLine();
// (6) Close the ServiceHostBase to shutdown the service.
selfHost.Close();
```

Without (4), the service will be .Net - .Net remoting!

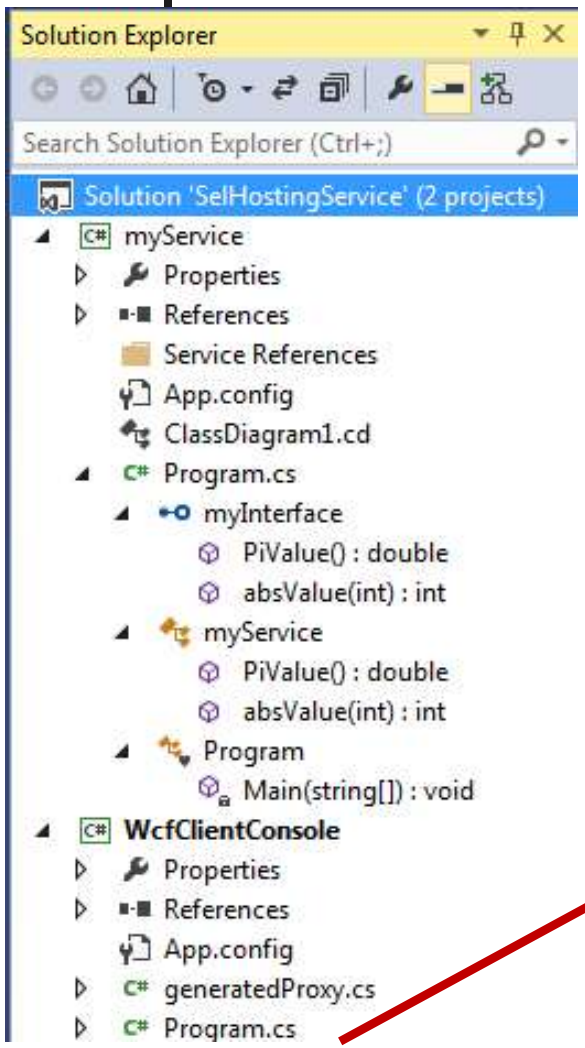
Starting the Service

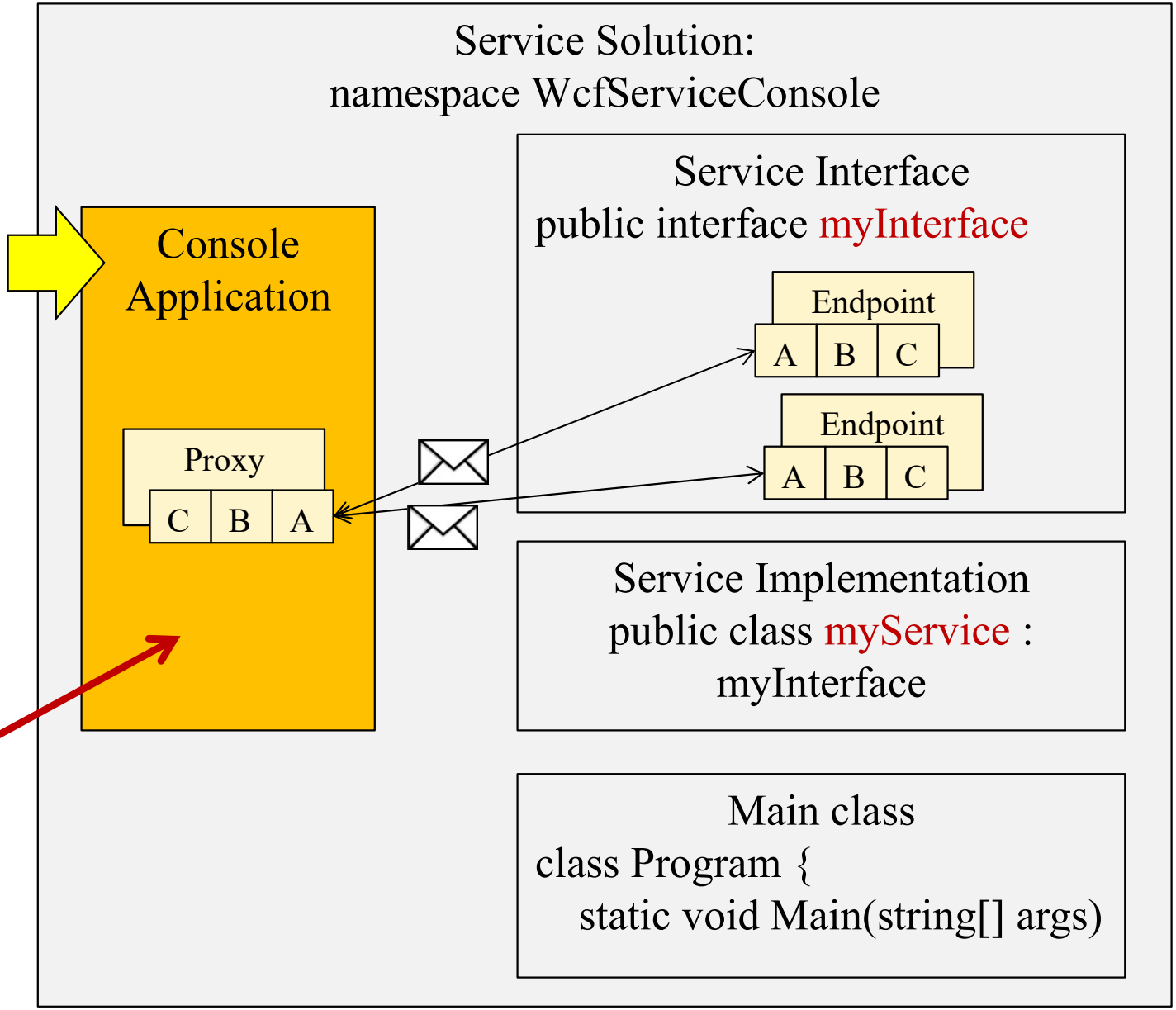
- The service is fully developed and we can start the service in Visual Studio by executing the command "Start without Debugging".



- You could also add and edit the endpoint by editing the Web.config file:
  http://msdn.microsoft.com/en-us/library/ms734786(v=vs.110).aspx

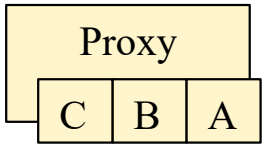
Project Overview: Service and Client





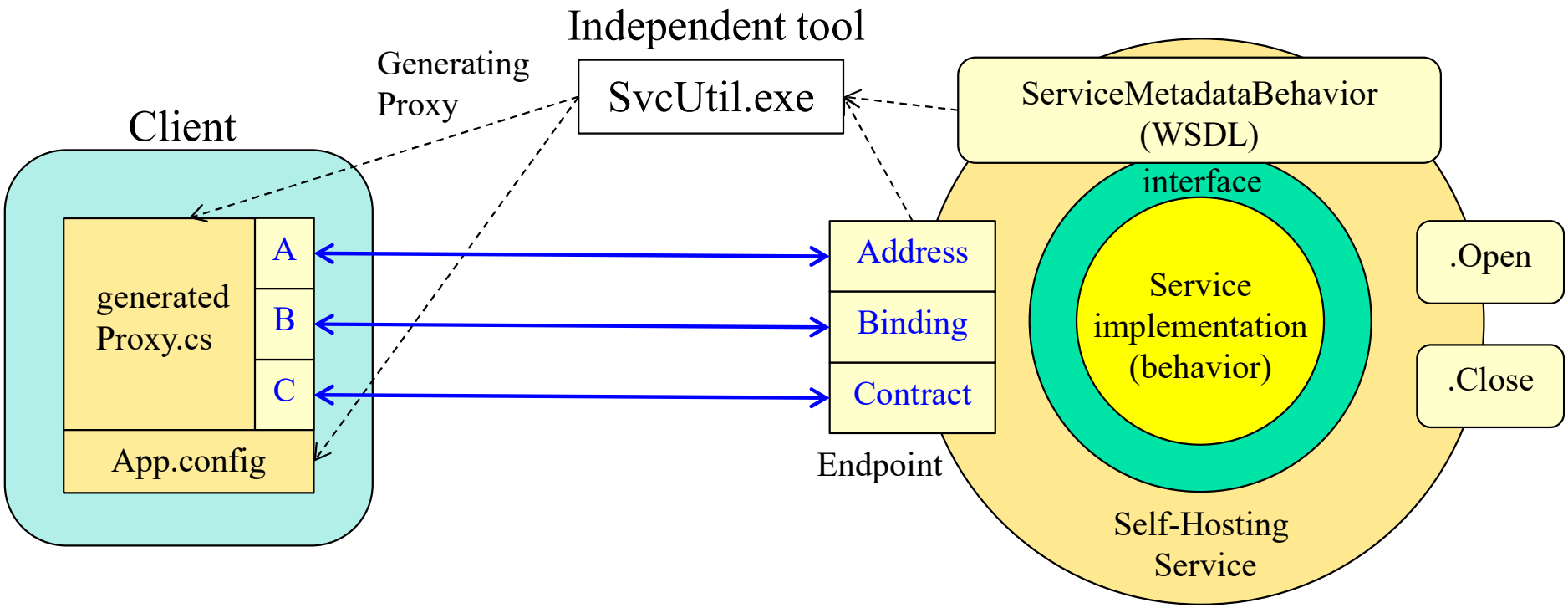
Client to Consume Self-Hosting Service in a Console Application

- Add a new Console Application in the same solution where the self-hosting project resides;
- Right-click to add Reference: System.ServiceModel under Assemblies → framework (It may have been added already in the newer version of VS). Another way is to find it with the NuGet package System.ServiceModel.Primitives
- Generate proxy to the service;
  - The service is not hosted in IIS nor in IIS Express server;
  - If WSDL is not generated in API, the proxy **cannot** be generated by using "Add Service Reference" in your programming environment.
  - Follow the next page to access the service:



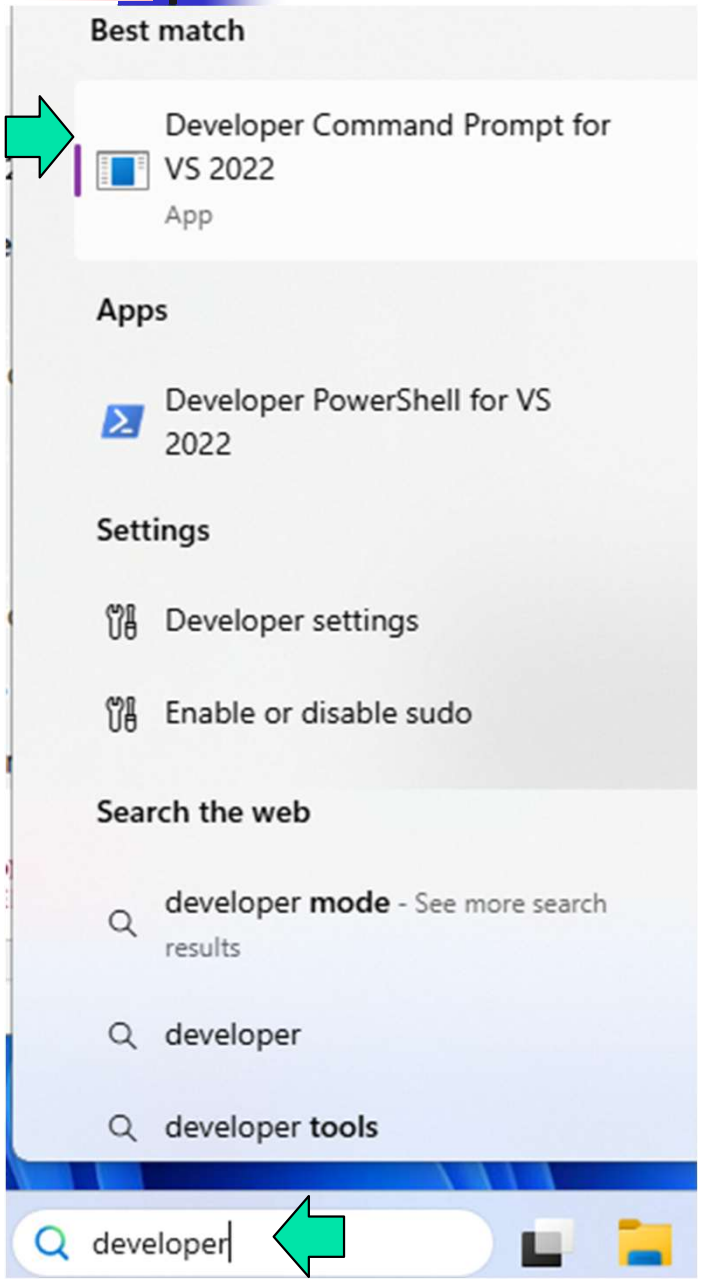
Creating the Proxy to Self-Hosting Service

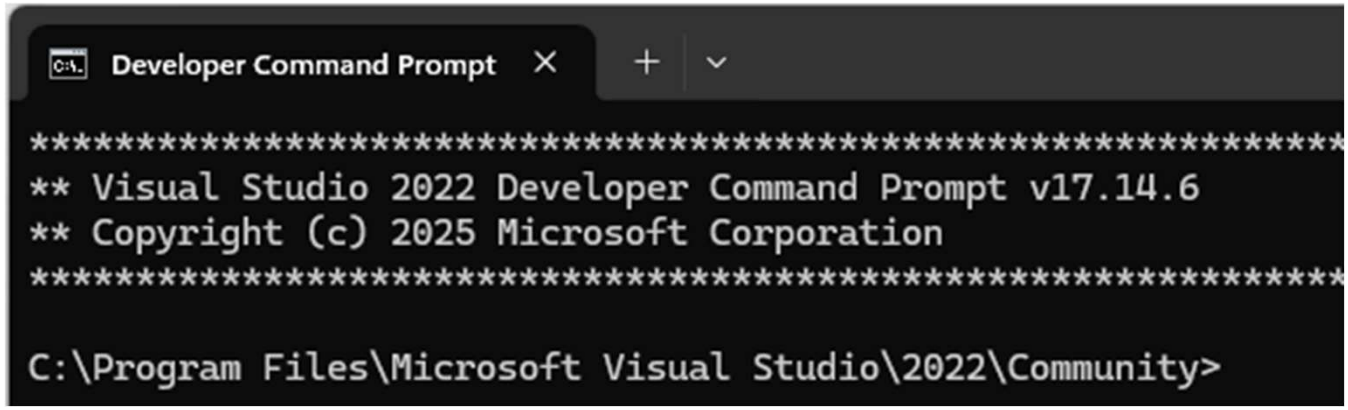
- Generate the proxy based on the metadata published by the service.
- Use the Service Model Metadata Utility Tool (SvcUtil.exe) tool to generate the proxy.



Use SvcUtil.exe Tool to Generate the Proxy

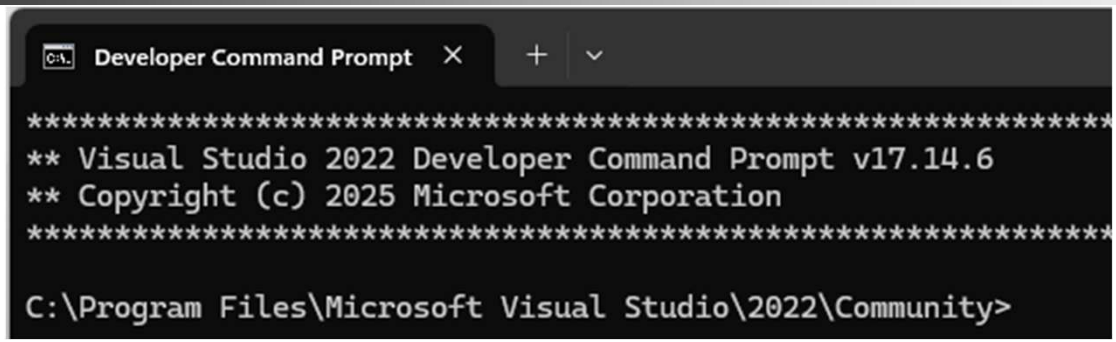
https://docs.microsoft.com/en-us/dotnet/framework/tools/developer-command-prompt-for-vs





Use SvcUtil.exe Tool to Generate the Proxy

https://docs.microsoft.com/en-us/dotnet/framework/tools/developer-command-prompt-for-vs



- Use DOS commands "dir", "cd .." and "cd folder" to navigate into the directory where you have your client project created in the previous step.
- While keeping your Service/API running, type the following command in the command window to execute:
- svcutil.exe /language:cs /out:generatedProxy.cs /config:App.config http://localhost:8000/Service
- This command will generate two files: App.config and generatedProxy.cs, and place them into the directory: C:\Windows\System32 or Search for them.

Use SvcUtil.exe Tool to Generate the Proxy

- Copy/Move the two files generatedProxy.cs and App.config into you client project directory;
- Add them into your client project: Right-click the client project in Solution Explorer, select Add → Existing Item. Browse to the location of generatedProxy.cs to add.
- Repeat the step above to add the App.config configuration file: By default, the Add Existing Item dialog box filters out all files with a .config extension. To see these files select All Files (*.*) in the file type.
- Also read: Developer Command Prompt for Visual Studio https://msdn.microsoft.com/en-us/library/ms229859(v=vs.110).aspx

Add the Two Generated Files into Client

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'SelHostingService' (2 p

myService

Properties

References

Service References

App.config

ClassDiagram1.cd

Program.cs

Build Solution Ctrl+Shift+B

Rebuild Solution

Clean Solution

Run Code Analysis on Solution Alt+F11

Batch Build...

Configuration Manager...

Add

Set StartUp Projects...

Add Solution to Source Control...

Paste Ctrl+V

Rename

Open Folder in File Explorer

Properties Alt+Enter

New Project...

Existing Project...

New Web Site...

Existing Web Site...

New Item... Ctrl+Shift+A

Existing Item... Shift+Alt+A

New Solution Folder

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'SelHostingService' (2 pr

myService

Properties

References

Service References

App.config

ClassDiagram1.cd

Program.cs

WcfClientConsole

Properties

References

App.config

generatedProxy.cs

Program.cs

New files added to the project

App.config File

```
<?xml version="1.0" encoding="utf-8"?>
<configuration>
  <system.serviceModel>
    <bindings>
      <wsHttpBinding>
        <binding name="WSHttpBinding_myInterface"
                 closeTimeout="00:01:00" openTimeout="00:01:00"
                 receiveTimeout="00:10:00" sendTimeout="00:01:00"
                 bypassProxyOnLocal="false" transactionFlow="false"
                 hostNameComparisonMode="StrongWildcard"
                 maxBufferPoolSize="524288"
                 maxReceivedMessageSize="65536"
                 messageEncoding="Text" textEncoding="utf-8"
                 useDefaultWebProxy="true"
                 allowCookies="false">
          <readerQuotas maxDepth="32" maxStringContentLength="8192"
                 maxArrayLength="16384" maxBytesPerRead="4096"
                 maxNameTableCharCount="16384" />
          <reliableSession ordered="true" inactivityTimeout="00:10:00“ enabled="false" />
```

Editing these default values

App.config File

```
      <security mode="Message">
        <transport clientCredentialType="Windows"
              proxyCredentialType="None" realm="" />
        <message clientCredentialType="Windows"
              negotiateServiceCredential="true"
         algorithmSuite="Default" establishSecurityContext="true" />
      </security>
    </binding>
  </wsHttpBinding>
</bindings>
<client>
  <endpoint address=  "http://localhost:8000/Service/myService"
              binding="wsHttpBinding" bindingConfiguration="WSHttpBinding_myInterface"
              contract="myInterface" name="WSHttpBinding_myInterface">
    <identity> <userPrincipalName value="YinongBYENG\yinong" /> </identity>
  </endpoint>
</client>
</system.serviceModel>
</configuration>
```

generatedProxy.cs File

```
public partial class myInterfaceClient : System.ServiceModel.ClientBase<myInterface>, myInterface
{

    1 reference
    public myInterfaceClient()
    {
    }

    0 references
    public myInterfaceClient(string endpointConfigurationName) :
            base(endpointConfigurationName)
    {
    }

    0 references
    public myInterfaceClient(string endpointConfigurationName, string remoteAddress) :
            base(endpointConfigurationName, remoteAddress)
    {
    }

    0 references
    public myInterfaceClient(string endpointConfigurationName, System.ServiceModel.EndpointAddress
            base(endpointConfigurationName, remoteAddress)
    {
    }

    0 references
    public myInterfaceClient(System.ServiceModel.Channels.Binding binding, System.ServiceModel.End
            base(binding, remoteAddress)
    {
    }
```

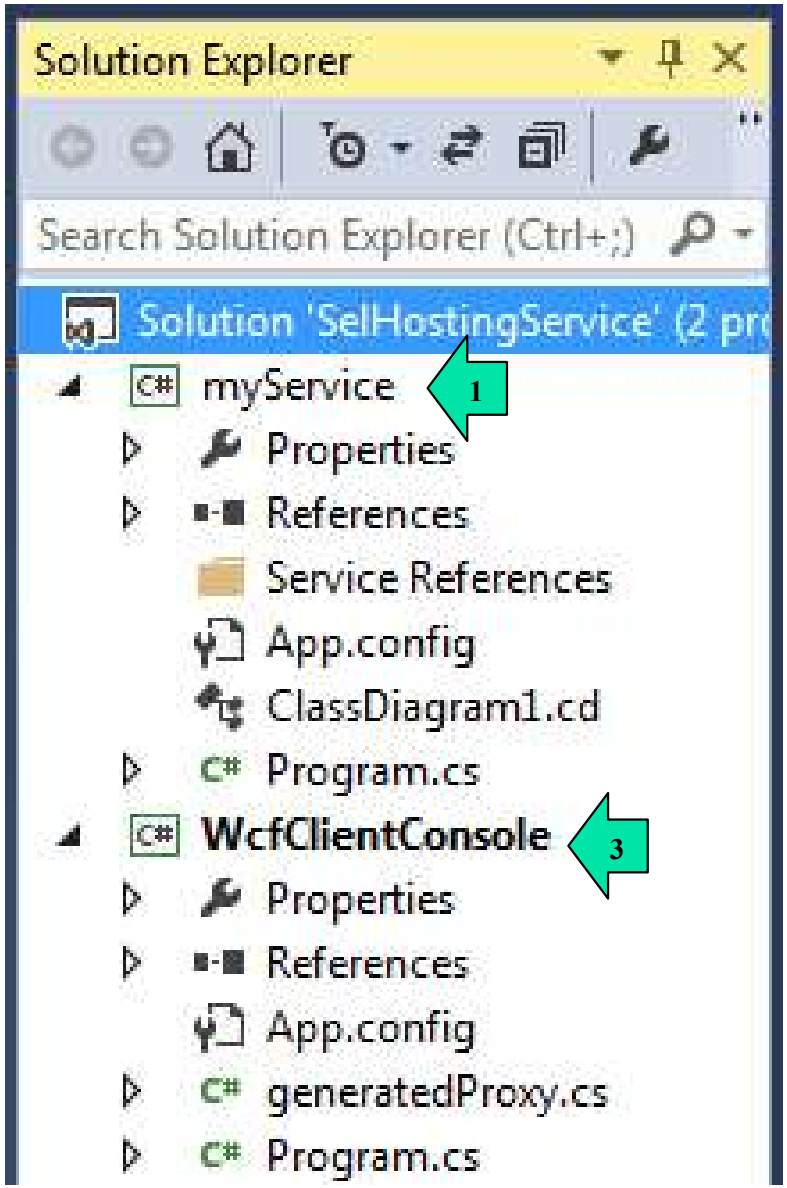
Client Program in Console Application

```
class Program {
    static void Main(string[] args) {
        //(1) Create a proxy to the self-hosting service.
        myInterfaceClient myProxy = new myInterfaceClient();
        // (2) Call the service operations through the proxy
        double pi = myProxy.PiValue();  // call PiValue operation
        Int32 test1 = 27;   // Create test input 1
        Int32 test2 = -132; // Create test input 2
        Int32 result1 = myProxy.absValue(test1); // call operation
        Int32 result2 = myProxy.absValue(test2); // call operation
        Console.WriteLine("PI value = {0}", pi);
        Console.WriteLine("Absolute values of {0} is {1} and of {2} is {3}",
                                                    test1, test2, result1, result2);
        myProxy.Close(); // (3) Close the proxy & channel to the service
        Console.WriteLine("/nPress <ENTER> to terminate the client./n");
    }
```

Test client and self-hosting service

1. Right click myService project and choose: Set as StartUp Project;
2. Start the project without debugging;
3. Right click the project WCFClientConsole and choose Debug → Start new instance.

If the console window closes at the end, you can add Console.ReadLine at the end of the client program to prevent the console to close.



Use Localhost and ASP .Net GUI Client to Consume Self-Hosting Services

The previous example showed to create the proxy manually, so that you have full control of the proxy. You can also create the proxy automatically using Visual Studio tool.

- Step 1. Add an ASP .Net Web Application project into the Self-Hosting Service Solution
  - Right-click the Solution WcfServiceConsole. Select Add → New → Web Application …
  - Select ASP .Net Web App and the project "AspSite"
- Start the Service in localhost or in IIS
- Add Service Reference
  - http://localhost:8000/Service/myService

Add Service Reference

Add Service Reference

To see a list of available services on a specific server, enter a service URL and click Go. To browse for available services, click Discover.

Address:

http://localhost:8000/Service

Go | Discover

Services:

- myService
  - myInterface

Operations:

- PiValue
- absValue

1 service(s) found at address 'http://localhost:8000/Service'.

Namespace:

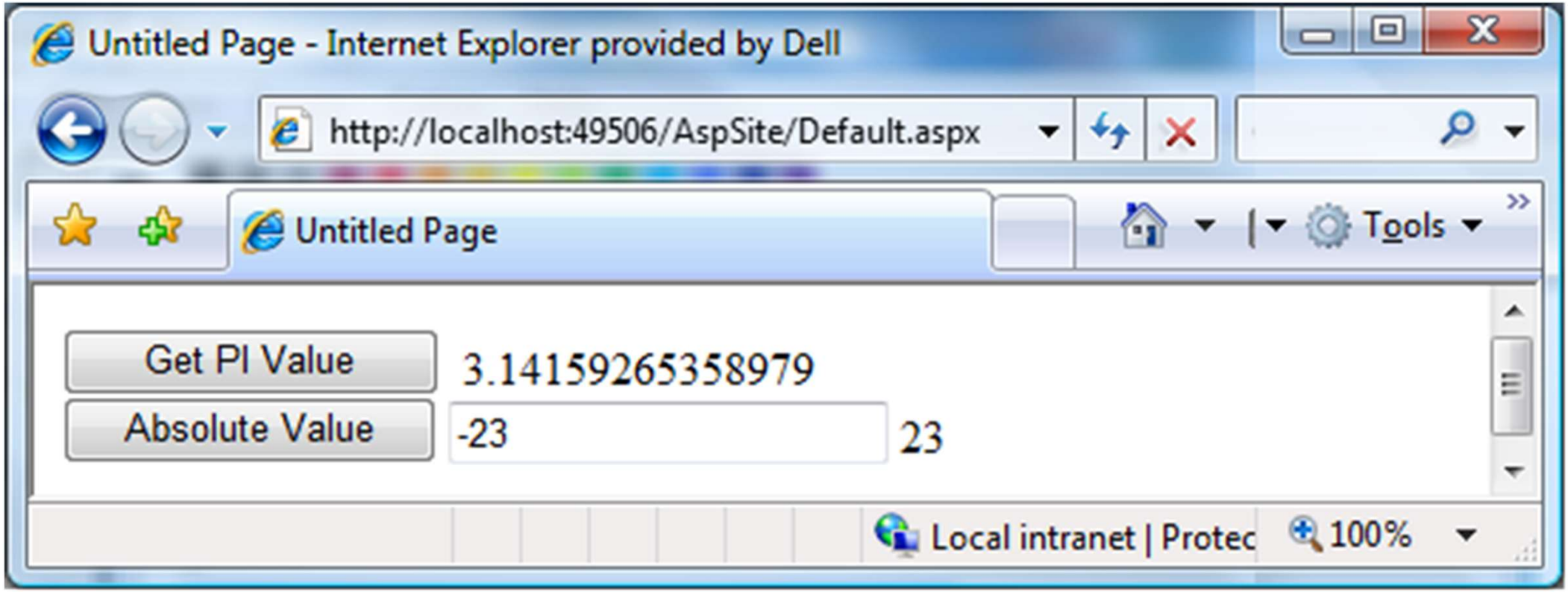
AspProxyToWcf

Advanced... | OK | Cancel

C# Code Behind the ASPX Page

```
using System;
public partial class _Default : System.Web.UI.Page {
    protected void Page_Load(object sender, EventArgs e) { }
    protected void btnPiValue_Click(object sender, EventArgs e) {
        AspProxyToWcf.myInterfaceClient refvar = new
                AspProxyToWcf.myInterfaceClient();
        double pi = refvar.PiValue();
        lblPiValue.Text = Convert.ToString(pi);
    }
    protected void Button1_Click(object sender, EventArgs e) {
        AspProxyToWcf.myInterfaceClient refvar = new
                AspProxyToWcf.myInterfaceClient();
        string abstxt = txtAbsoluteValue.Text;
        int absVal = refvar.absValue(Convert.ToInt32(abstxt));
        lblAbsoluteValue.Text = Convert.ToString(absVal);
    }
}
```

Use ASP .Net GUI Client to Consume Self-Hosting Services



Summary

- Developing Basic Service Using WCF Template and Server Hosting
- Developing API without Server Hosting: Self-Hosting
  - Writing a service/API without using WCF tools
  - Writing hosting service without using a server
  - Creating Proxy without using "Add Service Reference"
- Consuming WCF services
  - In Console Application
  - In ASP .Net
  - Many more: WF, BPEL, Cloud Computing, etc.